torch.nn.functional.embedding#

https://pytorch.org/docs/stable/generated/torch.nn.functional.embedding.html

Description:

Generate a simple lookup table that looks up embeddings in a fixed dictionary and size. This module is often used to retrieve word embeddings using indices. The input to the module is a list of indices, and the embedding matrix, and the output is the corresponding word embeddings. See torch.nn.Embedding for more details. Note that the analytical gradients of this function with respect to entries in weight at the row specified by padding_idx are expected to differ from the numerical ones.

Function Signature

```
torch.nn.functional.embedding(input, weight,
padding_idx=None, max_norm=None, norm_type=2.0,
scale_grad_by_freq=False, sparse=False)[source]#

Parameters

Return type

Shape:
```

Returns:

Tensor

Code Examples

```
>>> # a batch of 2 samples of 4 indices each
>>> input = torch.tensor([[1, 2, 4, 5], [4, 3, 2, 9]])
>>> # an embedding matrix containing 10 tensors of size 3
>>> embedding_matrix = torch.rand(10, 3)
>>> F.embedding(input, embedding_matrix)
tensor([[[ 0.8490,  0.9625,  0.6753],
         [ 0.9666,  0.7761,  0.6108],
         [ 0.6246,  0.9751,  0.3618],
         [ 0.4161,  0.2419,  0.7383]],

        [[ 0.6246,  0.9751,  0.3618],
         [ 0.0237,  0.7794,  0.0528],
         [ 0.9666,  0.7761,  0.6108],
         [ 0.3385,  0.8612,  0.1867]]])

>>> # example with padding_idx
>>> weights = torch.rand(10, 3)
>>> weights[0, :].zero_()
>>> embedding_matrix = weights
>>> input = torch.tensor([[0, 2, 0, 5]])
>>> F.embedding(input, embedding_matrix, padding_idx=0)
tensor([[[ 0.0000,  0.0000,  0.0000],
         [ 0.5609,  0.5384,  0.8720],
         [ 0.0000,  0.0000,  0.0000],
         [ 0.6262,  0.2438,  0.7471]]])
```

Notes & Warnings

NOTE Note Note that the analytical gradients of this function with respect to entries in weight at the row specified by padding_idx are expected to differ from the numerical ones.

NOTE Note Note that :class:`torch.nn.Embedding differs from this function in that it initializes the row of weight specified by padding_idx to all zeros on construction.

Detailed Documentation

Documentation

Generate a simple lookup table that looks up embeddings in a fixed dictionary and size.

This module is often used to retrieve word embeddings using indices. The input to the module is a list of indices, and the embedding matrix, and the output is the corresponding word embeddings.

See torch.nn.Embedding for more details.

Note that the analytical gradients of this function with respect to entries in weight at the row specified by padding_idx are expected to differ from the numerical ones.

Note that :class:`torch.nn.Embedding differs from this function in that it initializes the

row of weight specified by padding_idx to all zeros on construction.

input (LongTensor) – Tensor containing indices into the embedding matrix

weight (Tensor) – The embedding matrix with number of rows equal to the maximum possible index + 1, and number of columns equal to the embedding size

padding_idx (int, optional) – If specified, the entries at padding_idx do not contribute to the gradient; therefore, the embedding vector at padding_idx is not updated during training, i.e. it remains as a fixed “pad”.

max_norm (float, optional) – If given, each embedding vector with norm larger than max_norm is renormalized to have norm max_norm. Note: this will modify weight in-place.

norm_type (float, optional) – The p of the p-norm to compute for the max_norm option. Default 2.

scale_grad_by_freq (bool, optional) – If given, this will scale gradients by the inverse of frequency of the words in the mini-batch. Default False.

sparse (bool, optional) – If True, gradient w.r.t. weight will be a sparse tensor. See Notes under torch.nn.Embedding for more details regarding sparse gradients.

Input: LongTensor of arbitrary shape containing the indices to extract

Weight: Embedding matrix of floating point type with shape (V, embedding_dim), where V = maximum index + 1 and embedding_dim = the embedding size

Output: (*, embedding_dim), where * is the input shape